Fiscal Year:	FY 2006	**Task Last Updated:**	FY 08/24/2010
PI Name:	Plawsky, Joel Sc.D.		
Project Title:	An Experimental Study of Contact Line Dynamics on Nanostructured Surfaces		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Interfacial phenomena		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	12180-3522	**Congressional District:**	20
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03-C
Start Date:	09/01/2004	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Added to Task Book in 10/2006 per GRC		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Wayner, Peter (Rensselaer Polytechnic Institute)		
Grant/Contract No.:	NNC05GA41G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	We have discovered that evaporating menisci become unstable as the heat input into the fluid film increases. The menisci first undergo a transition to regular oscillations followed by a further transition to chaotic motion. This occurs whether the fluid is wetting or partially wetting. We propose to study the motion of these menisci using an improved version of our Image Analyzing Interferometry technique that allows us to measure the curvature, curvature gradient, and contact angle at the contact line and the adsorbed film thickness ahead of the contact line. In particular we are interested in the effect that surface structure has on the stability of these menisci. Three classes of structured surfaces will be studied; surfaces coated with a nanoporous silica layer, surfaces patterned with metallic nanorods, and surfaces patterned with carbon nanotubes. In each case the thickness of the surface texturing will be in the sub-200 nm range. Our primary goals for the work will be to determine the onset of instability, the oscillation frequency as a function of heat input, and the mechanism underlying the instability as a function of fluid and surface type. Three main fluids will		

	be studied; pentane, n-butanol, and water (with super-spreading surfactant). We will develop phenomenological, continuum based models to help explain and analyze the data.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of this research may effect the design of all heat transfer systems that rely on evaporation or condensation processes to transfer the heat from one location to another. The research aims to produce efficient heat transfer surfaces that would be used in heat pipes, in evaporation coils, and in dropwise condensation systems. The technology would be useful for cooling of microelectronic devices, for the cooling of photonic systems such as LEDs or laser diodes, and even for fabricating implantable systems for cooling or heating regions of the body.
Task Progress:	CONCLUSIONS Rough Surface Our experiments prove that the wetting properties in Isothermal and non-isothermal conditions are related to the surface features of a rough surface. Rough surfaces leads to enhanced wetting for a completely wetting fluid which is observed as thicker adsorbed films, lower tangent angle and lower peak curvature value in the transition region at all values of the heat flux. Increased wetting is a result of increase in surface potential that generates a higher disjoining pressure in the adsorbed film region. Length of the contact line increases due to roughness and therefore more area is available for evaporation. Two-scale roughness of deposited oxide further increases wetting. Higher recession on the rough surface indicates increased evaporation during unsteady state transition. During steady state the heat transfer model shows increased evaporation from the rough surface. Results obtained from the shear stress model support the presence of higher slip on rough surfaces. Nanoporous Surfaces A reflectivity technique was developed to study the evaporation of a hydrocarbon fluid (pentane) from NPS coating. During drying, the drying rate was critically dependent on the fraction of pores filled. The drying rate decreased till phi is 0.35, beyond which a relatively steady rate was observed. Application of a heat flux to the sidewall meniscus produced streaming and a reflectivity/interferometry based technique was developed to obtain the liquid meniscus thickness profiles for such a system. During streaming a pressure gradient is generated towards the adsorbed thin film which is due to a decrease in the adsorbed film thickness on application of a heat flux. The reflectivity/interferometry technique developed here could provide further details into the streaming behavior of pentane sidewall meniscus under non isothermal conditions. Drops The contact angle obtained for a nonane drop on a smooth surface was found to be higher than that obtained on a rough surface. Pinning was not observed on the evaporating drops on either surface since the contact angles remained constant. Lower wetting and increased surface area on the rough surface significantly increases the rate of evaporation throughout the heating process.
Bibliography Type:	Description: (Last Updated: 10/14/2025)
Articles in Peer-reviewed Journals	Ojha M, Chatterjee A, Dalakos G, Wayner PC, Plawsky JL. "Role of solid surface structure on evaporative phase change from a completely wetting corner meniscus." Physics of Fluids 2010 May;22(5):052101. http://dx.doi.org./10.1063/1.3392771 , May-2010
Articles in Peer-reviewed Journals	Ojha M, Chatterjee A, Mont F, Schubert EF, Wayner PC Jr, Plawsky JL. "The role of solid surface structure on dropwise phase change processes." International Journal of Heat and Mass Transfer, 2010 Feb;53(5-6):910-22. http://dx.doi.org/10.1016/j.ijheatmasstransfer.2009.11.033 , Feb-2010
Articles in Peer-reviewed Journals	Plawsky JL, Ojha M, Chatterjee A, Wayner PC Jr. "Review of the effects of surface topography, surface chemistry, and fluid physics on evaporation at the contact line." Chemical Engineering Communications, 2009 May;196(5): 658-96. http://dx.doi.org/10.1080/00986440802569679 , May-2009
Books/Book Chapters	Ojha M, Dalakos G, Panchamgam S, Wayner PC Jr, Plawsky JL. "Effects of surface structure on the behavior of a heated contact line." in "Contact Angle, Wettability and Adhesion. Volume 5." Ed. K.L. Mittal. Leiden ; Boston : VSP, 2008. p. 3-24., Jan-2008
Papers from Meeting Proceedings	Plawsky JL, Ojha M, Chatterjee A, Wayner PC. "Enhancing Transport Processes by Controlling Contact Line Dynamics." Presented at ISTP-18 (International Symposium on Transport Phenomena), Seoul, Korea, August 2007. Proceedings ISTP-18 (International Symposium on Transport Phenomena), Seoul, Korea, August 2007. , Aug-2007
Papers from Meeting Proceedings	Ojha M, Plawsky JL, Wayner PC. "Reflectivity Based Study of Pentane Evaporation from a Porous Nano-structured Surface." Presented at the 13th International Heat Transfer Conference, Sydney, Australia, August 13-18, 2006. Proceedings of the International Heat Transfer Conference, Sydney, Australia, August 2006. , Aug-2006